Figure 13.3 In Grafana, you can browse and search log messages aggregated and stored by Loki.

When you're done testing the logging setup, stop all containers with `docker-compose down`.

NOTE Following the same approach, update the Docker Compose configuration for all the other Spring Boot applications in the Polar Bookshop system to use the Fluentd logging driver and rely on Fluent Bit for collecting logs. As a reference, you can look at the source code repository accompanying this book (Chapter13/13-end/polar-deployment/docker).

Logs provide some information about how an application behaves, but they're not enough to infer its internal state. The next section will cover how you can make applications expose more data about their health status.

13.2 Health probes with Spring Boot Actuator and Kubernetes

Once an application is deployed, how can we tell if it's healthy? Is it capable of handling new requests? Did it enter a faulty state? Cloud native applications should provide information about their health so that monitoring tools and deployment platforms can detect when there's something wrong and act accordingly. We need dedicated health endpoints to check on the status of the application and any components or services it might use.

The deployment platform can periodically invoke health endpoints exposed by applications. A monitoring tool could trigger an alert or a notification when an application instance is unhealthy. In the case of Kubernetes, the platform will check the health endpoints and automatically replace the faulty instance or temporarily stop sending traffic to it until it's ready to handle new requests again.